



Documented Dragon-King Events in the Tail of the Gutenberg-Richter Distribution

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广东·深圳

Introduction

The outstanding question we discuss in this talk is **whether large earthquakes enjoy some distinguishable properties that could make them predictable**. This question is motivated both by *the quest of understanding the physics of earthquakes and their spatio-temporal organization* and by *the profound implications this would have for seismic risk assessment and mitigation*.

本报告想要讨论的主要问题是，大地震是否具有某些可区分的特性，从而使其可被预测。对于这个问题的探讨不仅有利于对地震物理及其时空组织结构的更好理解，而且对于地震灾害风险评估和防灾减灾实践也具有重要意义。

Introduction

(1) Viewpoint: No!

There is a widespread view that the answer to this question is negative: “**large earthquakes are just like small earthquakes that continued to grow to large sizes.**” In this picture, large earthquakes are not different from their smaller siblings, except for their sizes, with scaling relations describing the statistical properties of the full population of earthquakes. This perspective is informed by the many power law distributions characterizing seismicity statistics.

普遍的观点认为，这个问题的答案是否定的，即大地震仅仅是小地震继续扩展到更大尺度的结果。换句话说，除了大小不同，大地震与小地震并无任何差别，两者的统计特性也可以通过刻画地震总体特征的标度关系来描述。这一观点来源于许多描述地震活动统计特性的幂律分布。

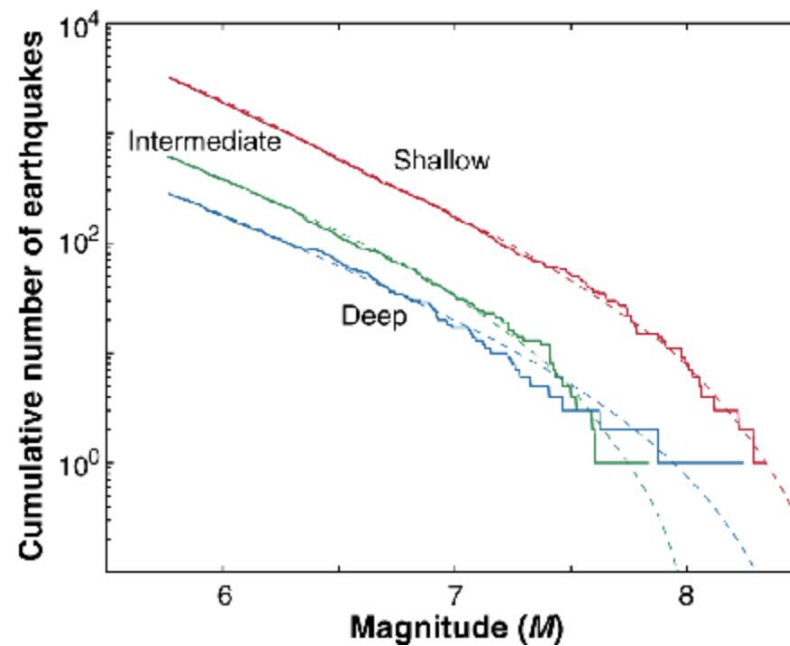
Introduction

(1) Viewpoint: No!

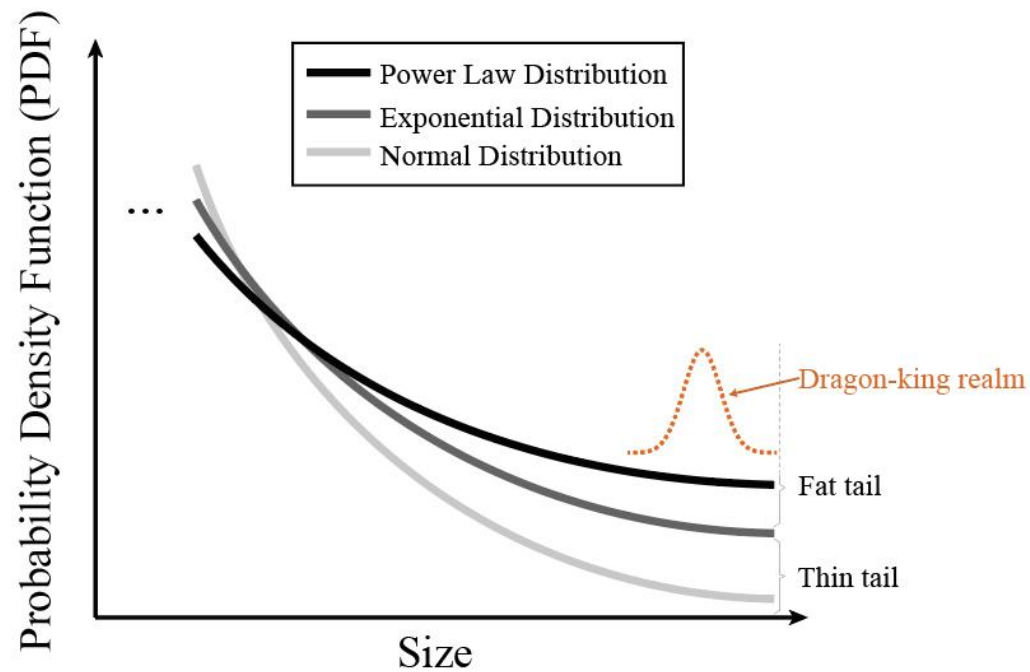
P.D.F. of power law $P(x) \propto \frac{1}{x^\alpha}$

Two key properties

- self-similarity (自相似性)
- fat tails (肥尾)



(Geller et al., 1997)



Introduction

(1) Viewpoint: No!

$$\text{P.D.F. of power law } P(x) \propto \frac{1}{x^\alpha}$$

Power law distributions that approximately describe many observations in seismology:

- [Gutenberg-Richter law](#) (Gutenberg and Richter, 1944; Sornette et al., 1996; Liu et al., 2024),
- [frequency distribution of fault rupture scales](#) (Davy, 1993; Scholz et al., 1993; Scholz, 1997; Clark et al., 1999),
- [the fractal/multifractal/hierarchical scaling features of seismicity and fault systems](#) (Hirata and Imoto, 1991; Turcotte, 1997; Hirabayashi et al., 1993; Ouillon et al., 1996),
- [the frequency distribution of seismicity rates in spatiotemporal grids](#) (Saichev and Sornette, 2006; 2007),
- [the frequency distribution of the time intervals between large earthquakes and their epicentral distances](#) (Ito, 1995), and
- [the frequency distribution of earthquake fault slip](#) (Lavallée and Archuleta, 2003; Lavallée et al., 2006), among others.

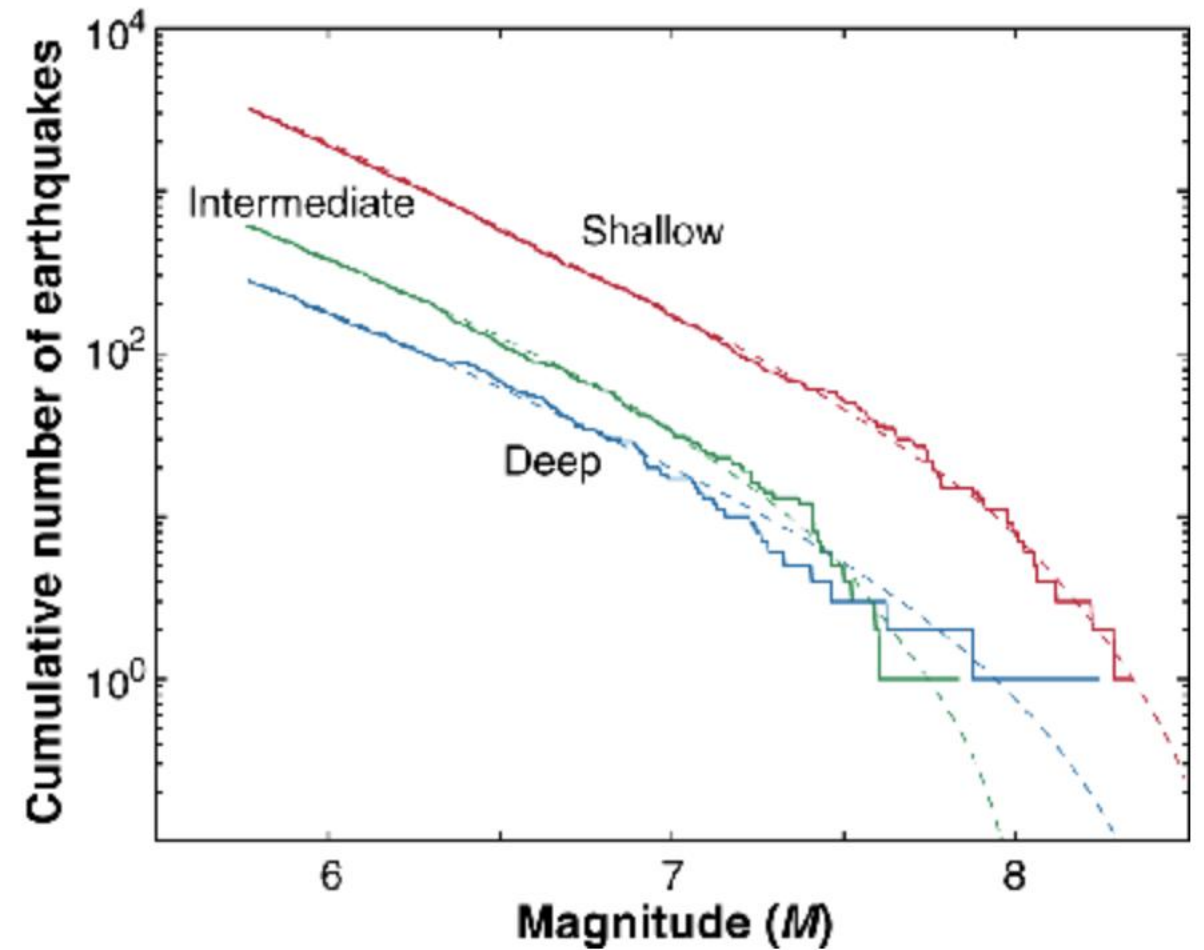
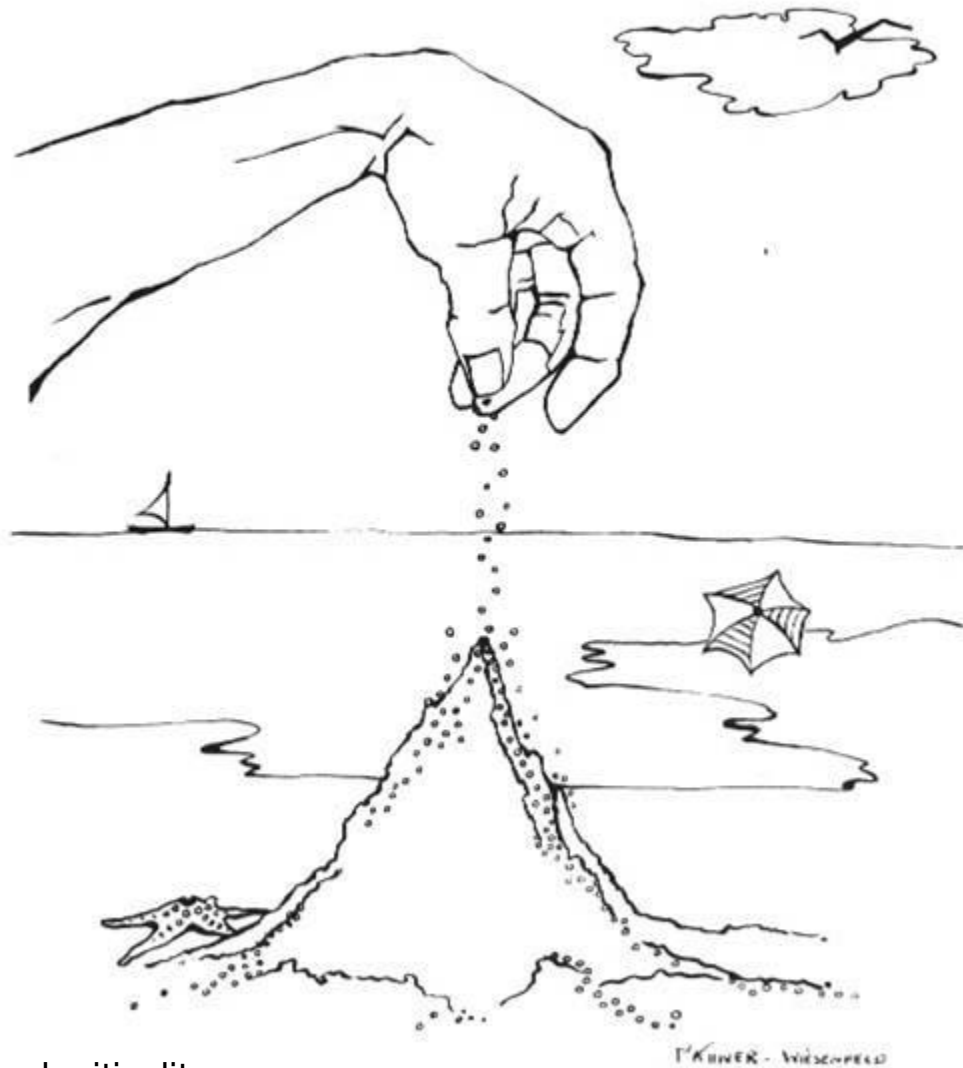
The evidence for self-similarity between earthquakes is further reinforced by **the existence of several approximate seismicity laws also formulated as power laws**

- [the Omori-Utsu law](#) describes the decay rate of aftershock activity following a mainshock (Omori, 1894; Utsu, 1970),
- [the fertility law](#) quantifies the exponential dependence of the average number of aftershocks triggered by a given mainshock as a function of its magnitude (it is thus a power law function of the earthquake energy) (Yamanaka and Shimazaki, 1990),
- [the spatial Green function](#) specifies the decay of the number of aftershocks with distance from the mainshock's epicenter (Zhuang et al., 2004; Nandan et al., 2021), and
- [the \$k^{-2}\$ spectrum in the wavenumber domain](#) describes the relative content of spatial frequency components of earthquake fault slips (Herrero and Bernard, 1994), among others.

Introduction

(1) Viewpoint: No!

P.D.F. of power law $P(x) \propto \frac{1}{x^\alpha}$



(Geller et al., 1997)

Introduction

(1) Viewpoint: No!

P.D.F. of power law $P(x) \propto \frac{1}{x^\alpha}$

Earthquakes Cannot Be Predicted

ROBERT J. GELLER, DAVID D. JACKSON, YAN Y. KAGAN, AND FRANCESCO MULARGIA [Authors Info & Affiliations](#)

SCIENCE • 14 Mar 1997 • Vol 275, Issue 5306 • p. 1616 • DOI: 10.1126/science.275.5306.1616

Abstract

Can the time, location, and magnitude of future earthquakes be predicted reliably and accurately? In their Perspective, Geller *et al.*'s answer is “no.” Citing recent results from the physics of nonlinear systems “chaos theory,” they argue that any small earthquake has some chance of cascading into a large event. According to research cited by the authors, whether or not this happens depends on unmeasurably fine details of conditions in Earth's interior. Earthquakes are therefore inherently unpredictable. Geller *et al.* suggest that controversy over prediction lingers because prediction claims are not stated as objectively testable scientific hypotheses, and due to overly optimistic reports in the mass media.

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Cannot Earthquakes Be Predicted?

MAX WYSS [Authors Info & Affiliations](#)

SCIENCE • 17 Oct 1997 • Vol 278, Issue 5337 • pp. 487-490 • DOI: 10.1126/science.278.5337.487

TECHNICAL COMMENTS

Cannot Earthquakes Be Predicted?

Robert J. Geller *et al.* are on shaky ground when they state, in the title of their Perspective (1), that earthquakes cannot be predicted. In spite of their advice, we should not stop studying the physics of preparation for catastrophic rupture, in the field, the laboratory, and theoretically; neither should we stop measuring crustal na-

ic failure in a major earthquake. This phenomenon is routinely observed in the laboratory in stiff rock presses, and it has been modeled quantitatively by modern friction laws (16), with the result that years before large subduction shocks occur, a reduction of stress is expected near the source volume (17). Thus, several types of measurements

however, large earthquakes cannot occur, regardless of the nature of the crack distribution. Along the plate margins, where the vast majority of all earthquakes occur, the stresses are likely to be low, and the stress release may be nearly complete (24).

At the time of Columbus, most experts asserted that one could not reach India by sailing from Europe to the west and that funds should not be wasted on such a folly. Geller *et al.* make a similar mistake, but I doubt that human curiosity and ingenuity

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Introduction

(1) Viewpoint: No!

P.D.F. of power law $P(x) \propto \frac{1}{x^\alpha}$

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Debate in 1999

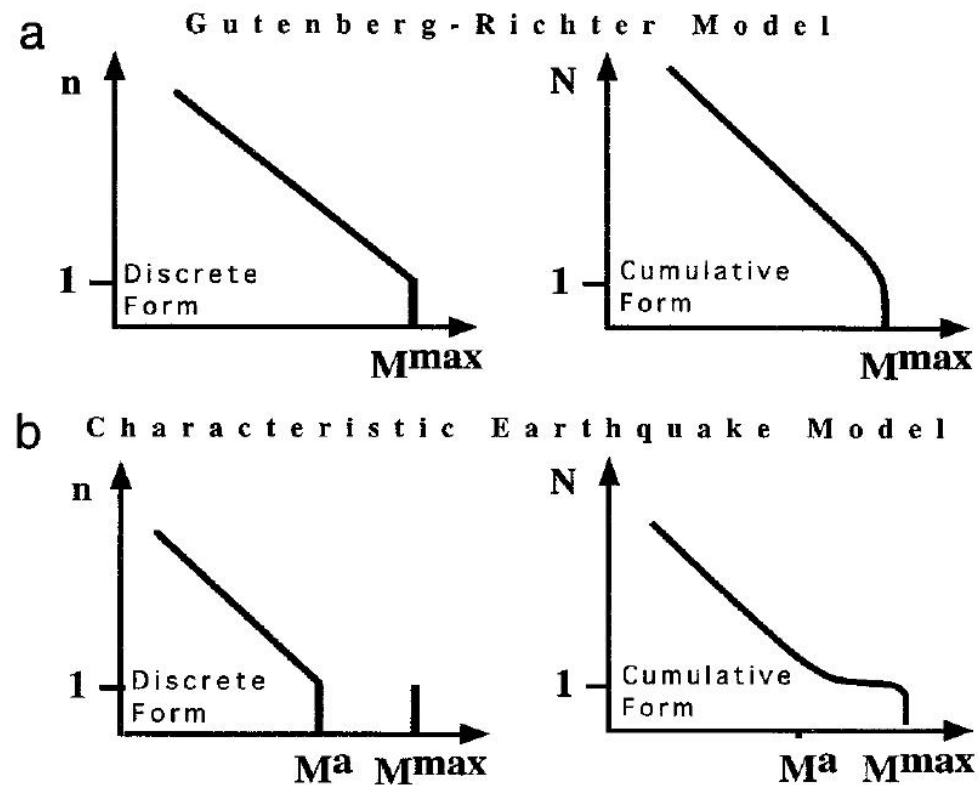
Article Type	Year
Debate (67)	1999 (67)
Debate 08 Apr 1999 <u>Earthquake debate - Didier Sornette</u> Predicting earthquakes requires an understanding of the underlying physics, which calls for novel multidisciplinary approaches at a level never yet undertaken. Notwithstanding past efforts in several countries in the last decades, I fail to see that the scientific community has used the full potential of artificial/computational intelligence, statistical physics, super-computer modelling, large scale monitoring of a full spectrum of physical measurements, coupled together with more traditional seismological and geological approaches to make a dent in the earthquake problem. What we have learned is that past failures in earthquake prediction reflect the biased view that it was a simple problem. Didier Sornette	
Debate 08 Apr 1999 <u>Earthquake prediction: Concluding Remarks</u> This debate has highlighted both a degree of consensus and a degree of continuing controversy within the thorny subject of the predictability of earthquakes. In terms of the four levels of prediction of seismicity I introduced at the start of this debate, a consensus has emerged that at least some form of time-dependent seismic hazard can be justified on both physical and observational grounds. The phenomenon of earthquake triggering leads to a transient, local increase in probability of future earthquakes, for example as aftershocks, but also sometimes in the form of subsequently larger events. In fact warnings based on such clustering are already in use in California (Michael, week 2). On the other hand, all of the contributors to this debate who expressed an opinion agree that the deterministic prediction of an individual earthquake, within sufficiently narrow limits to allow a planned evacuation programme, is an unrealistic goal. Ian Main	
Debate 01 Jan 1999 <u>Earthquake debate - Christopher Scholz</u> Geller and Jackson have both reproached me for not citing all of the Jackson and Kagan papers in my earlier statement. Space requirements did not allow for a fuller discussion it that time. Christopher Scholz	

(<https://www.nature.com/nature/articles?searchType=journalSearch&sort=PubDate&type=debate&year=1999&page=4>. Last accessed March 28, 2024.)

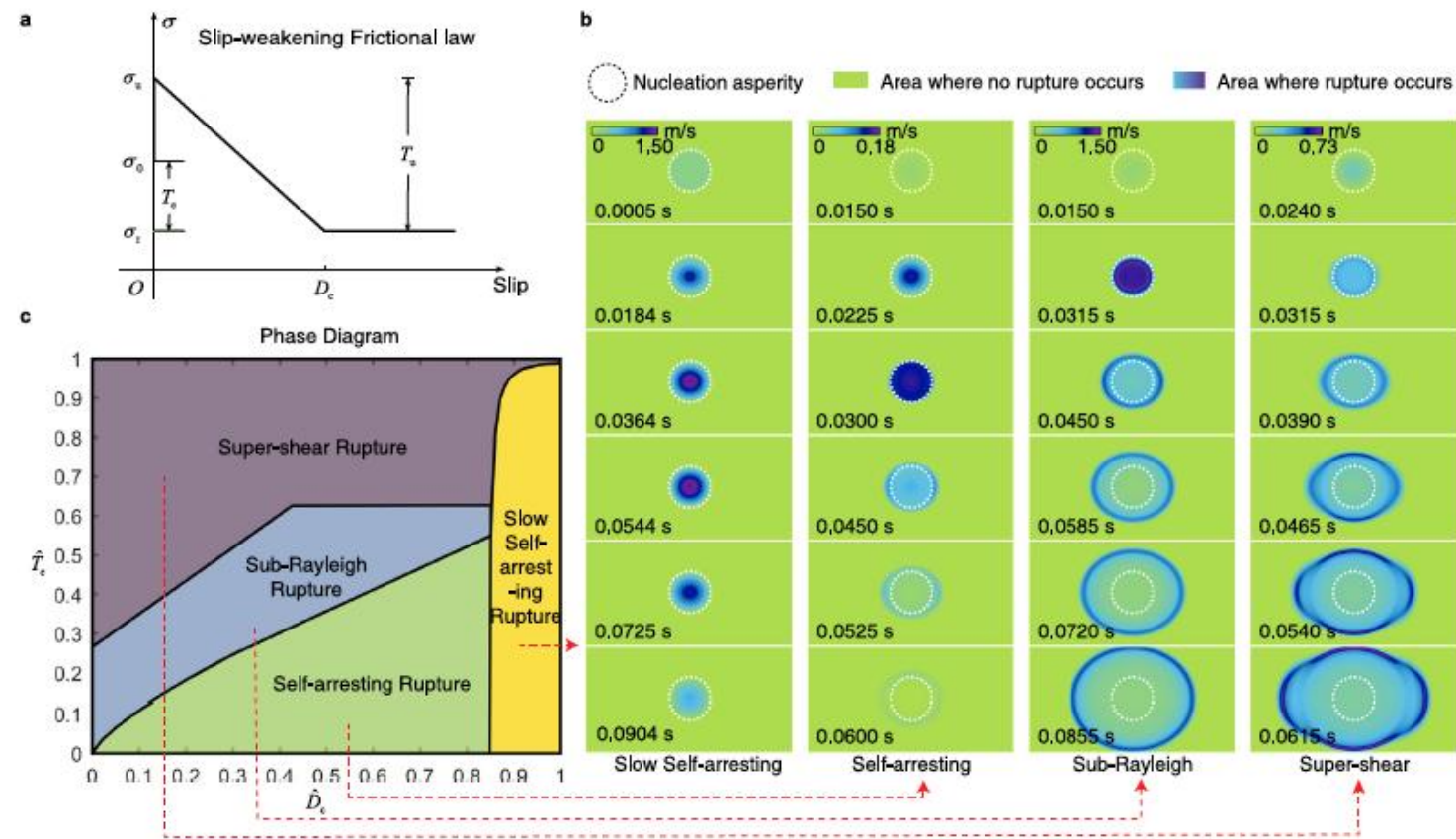
Debate 25 Mar 1999	<u>Earthquake debate - Max Wyss</u> Max Wyss
Debate 25 Mar 1999	<u>Earthquake debate - Zhongliang Wu</u> Testing hypotheses is an essential part of earthquake prediction. The 'game rule' associated with this test is especially important because it leads to the criteria for accepting or rejecting a statistical hypothesis related to earthquake prediction. Zhongliang Wu
Debate 25 Mar 1999	<u>Stress-forecasting: an apparently viable third strategy</u> All discussions so far have referred (perhaps not surprisingly) to the properties of earthquakes, their times, locations, nucleation mechanisms, physics of the source, possible precursors, etc. I think this will lead nowhere. Earthquakes are extraordinarily varied and impossible to average. Perhaps the only feature of earthquakes that can be relied on is that they release a large amount of stress which, because rock is weak, has necessarily accumulated over a large volume of rock. If this build up of stress can be monitored then the time and magnitude of the earthquake when fracture criticality is reached can be subjected to 'stress-forecast'. I suggest that we already know how to do this. The effects have been seen with hindsight for eight earthquakes worldwide, and the time and magnitude of an M=5 earthquake has been successfully stress-forecasted. Stuart Crampin
Debate 18 Mar 1999	<u>Earthquake debate - David Jackson</u> Scholz omitted crucial parts of the recent history of the seismic gap forecast and test. He remarked that our test of the gap theory was 'flawed' because it used earthquakes 'smaller than system-sized'. This was also asserted in a published comment by Nishenko & Sykes ¹ and answered by Jackson & Kagan ² . But 'system-sized' was never defined in the original seismic gap model ³ . David Jackson
Debate 18 Mar 1999	<u>The status of earthquake prediction</u> David D. Jackson
Debate 18 Mar 1999	<u>On the existence and complexity of empirical precursors</u> Earthquake prediction is strictly related to empirical precursors. Despite the results presented in recent decades in support of the existence of empirical precursors there is scepticism in the scientific community about whether they exist ¹⁻³ . The widespread argument is that precursor signals reported are unrelated to earthquake activity and that they could have occurred by chance. If this were true, earthquake prediction would not be possible. Francesco Biagi

Introduction

(1) Viewpoint: Yes!



(Wesnousky, 1994; Ishibe and Shimazaki, 2012)



(Wen et al., 2018; Wei et al., 2021; Xu et al., 2023; Sornette et al., 2024)

Some important animals in the history of human science



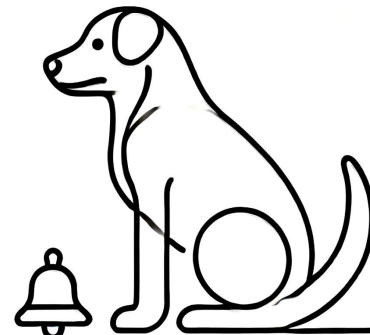
Taleb's Black Swan
(extreme risks of unknown unknowns)



Walker's Grey Rhino
(high-probability extreme risks that have been ignored)



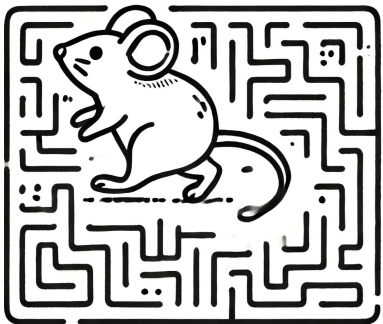
Sornette's Dragon-king
(extreme events that differ from those within the same system)



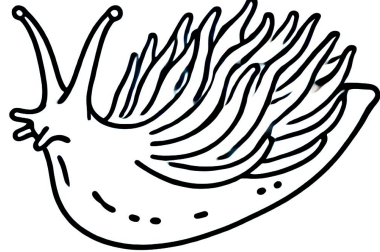
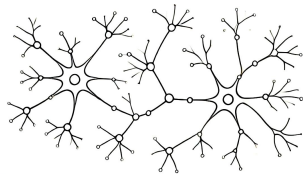
Pavlov's Dog
(conditioned reflex)



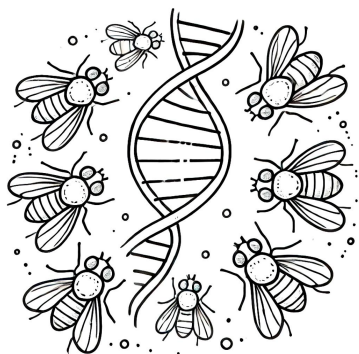
Spallanzani's Eagle
(digestive enzymes)



Tolman's Rat
(maze learning)



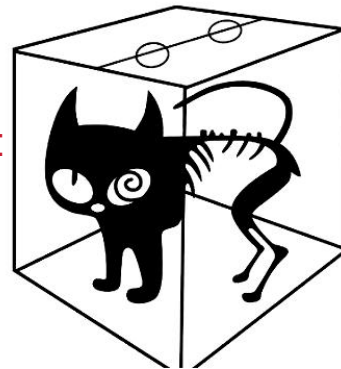
Kandel's Sea Slug
(long-term memory)



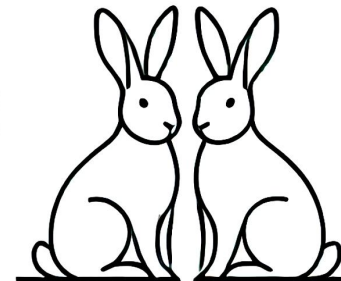
Morgan's Fruit Fly
(genetics)



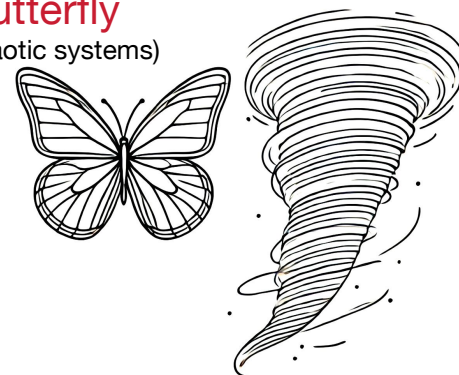
Zhuangzi's Butterfly
(reality and illusion)



Schrödinger's Cat
(quantum superposition)



Fibonacci's Rabbits
(golden ratio sequence)



Lorenz's Butterfly
(butterfly effect in chaotic systems)

Some pics are generated by ChatGPT, while others are sourced from the internet

1 Dragon-king events and dragon-king theory

1.1 Definition of dragon-king events

Dragon-kings (DK) embody a dual metaphor, expressing that an event is both exceptionally large ("king") and originates from unique circumstances ("dragon") compared to its peers (Sornette, 2009; Sornette and Ouillon, 2012; de S. Cavalcante et al., 2013).

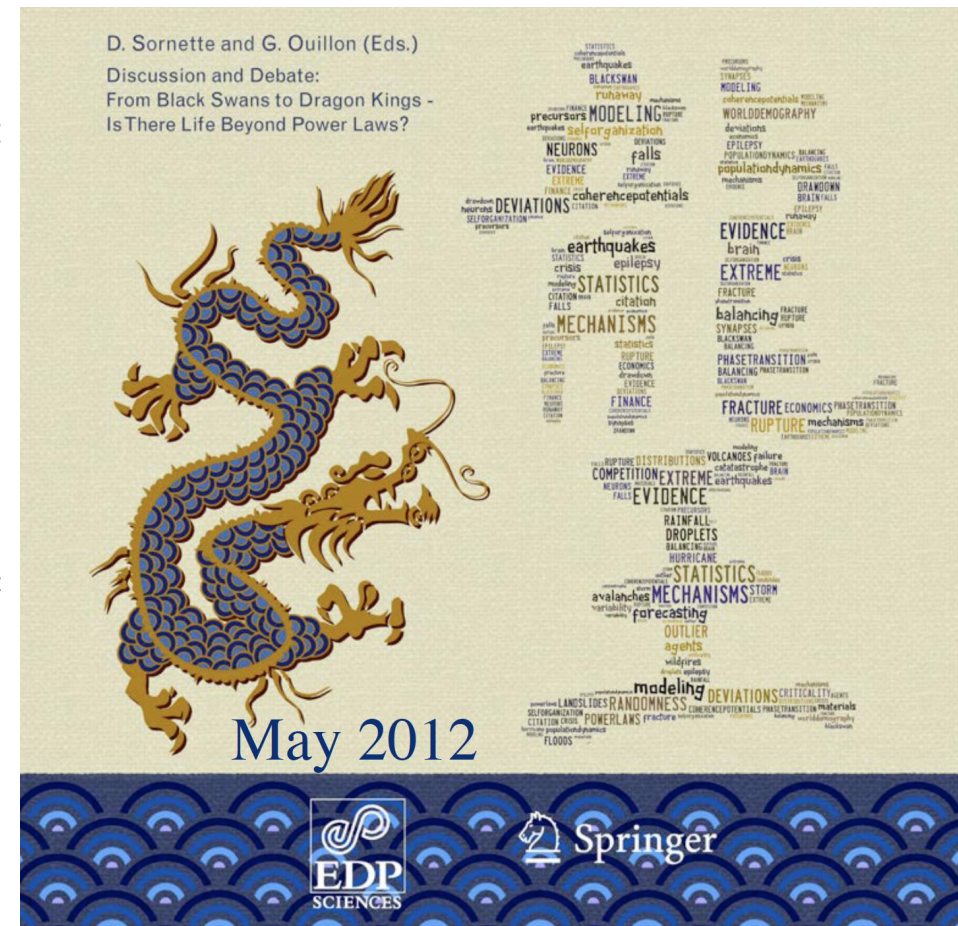
“龙王”体现了一种双重隐喻，表示一类不仅相较同一系统其他事件具有极大规模或影响力（因此称“王”），而且还具有独特起源环境（因此称“龙”）的极端事件 (Sornette, 2009; Sornette and Ouillon, 2012; de S. Cavalcante et al., 2013)。

Dragon-king events have been widely reported in many studies, including:

- 1) **Natural sciences**, such as material failure, earthquake ruptures (Sammis and Sornette, 2002), fluid dynamics (L'vov et al., 2001), landslide prediction (Lei et al., 2023), epileptic seizures (Osorio et al., 2010; de Arcangelis, 2012);
- 2) **Finance**, such as stock market crashes (Johansen and Sornette, 1998; 2001; 2010; Filimonov and Sornette, 2015);
- 3) **Sociology**, such as the impact of academic papers (Golosovsky and Solomon, 2012), city sizes (Wheatley and Sornette, 2015), and nuclear safety accidents (Wheatley et al., 2017), among others.

目前，“龙王”事件已在许多研究中被广泛报道，包括

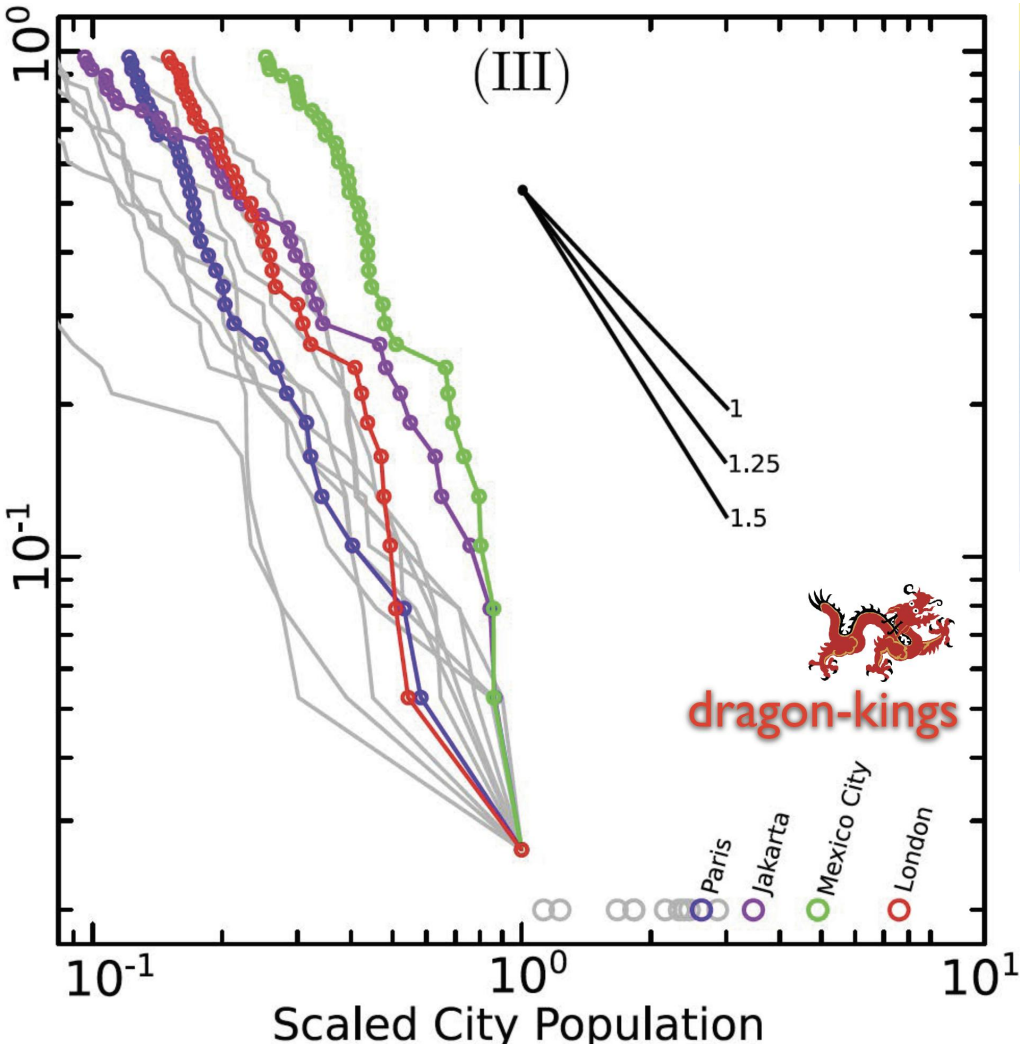
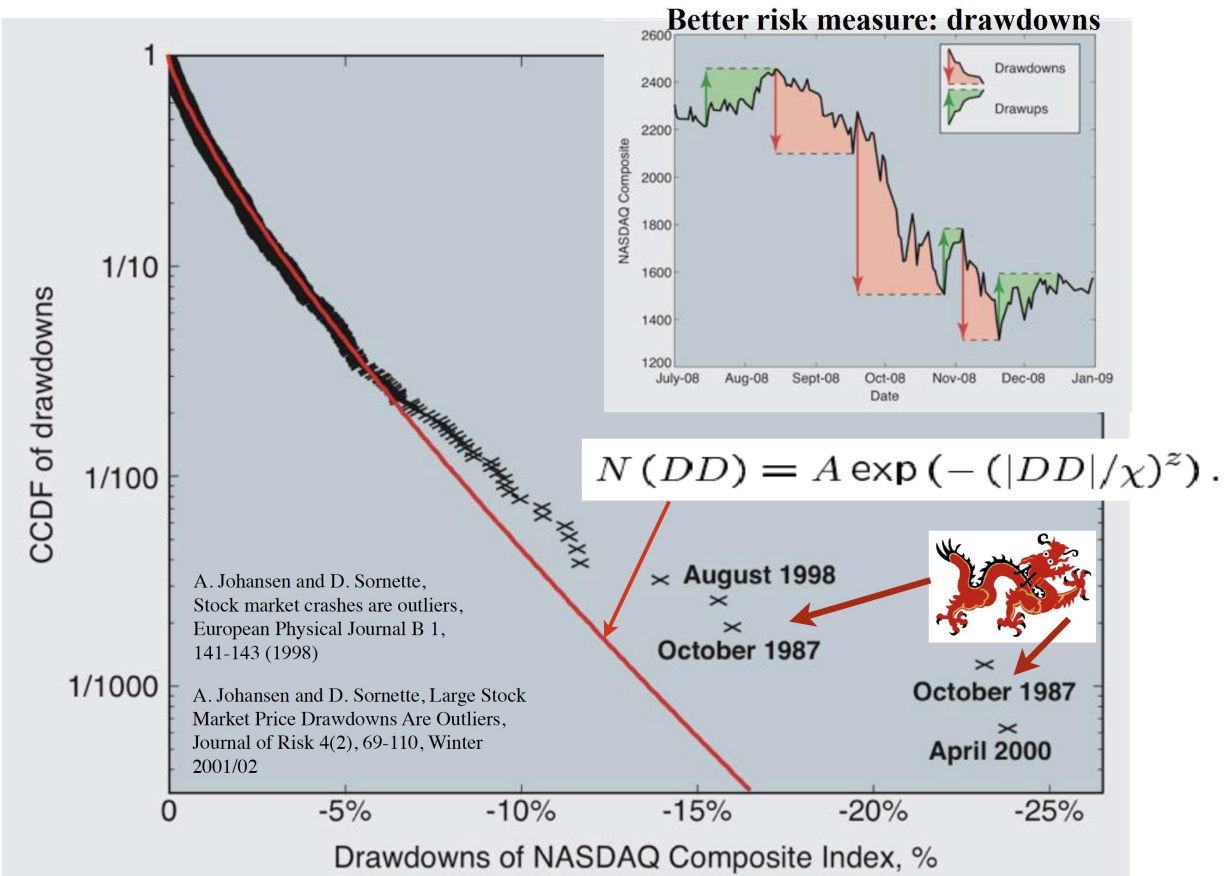
- 1) **自然科学**，如材料失效、地震破裂 (Sammis and Sornette, 2002)、流体力学 (L'vov et al., 2001)、滑坡预测 (Lei et al., 2023)、癫痫发作 (Osorio et al., 2010; de Arcangelis, 2012)；
- 2) **金融学**，如股市崩盘 (Johansen and Sornette, 1998; 2001; 2010; Filimonov and Sornette, 2015)；
- 3) **社会学**，如学术论文影响力 (Golosovsky and Solomon, 2012)、城市规模 (Wheatley and Sornette, 2015)、核安全事故 (Wheatley et al., 2017) 等；



1 Dragon-king events and dragon-king theory

1.1 Definition of dragon-king events

From outlier to dragon-king



1 Dragon-king events and dragon-king theory

1.2 Potential mechanisms of dragon-king events

Dragon-king events arise from unique amplification mechanisms that do not affect (or affect to lesser degree) other events in the same system, thus endowing dragon-king events with their inherent specific properties.

“龙王”事件的发生基于其独有的放大机制，该机制无法（或较少）作用于同一系统其他事件，因此赋予了“龙王”事件本质上的特殊属性。

- (1) Exogenous shocks
外源性冲击



1 Dragon-king events and dragon-king theory

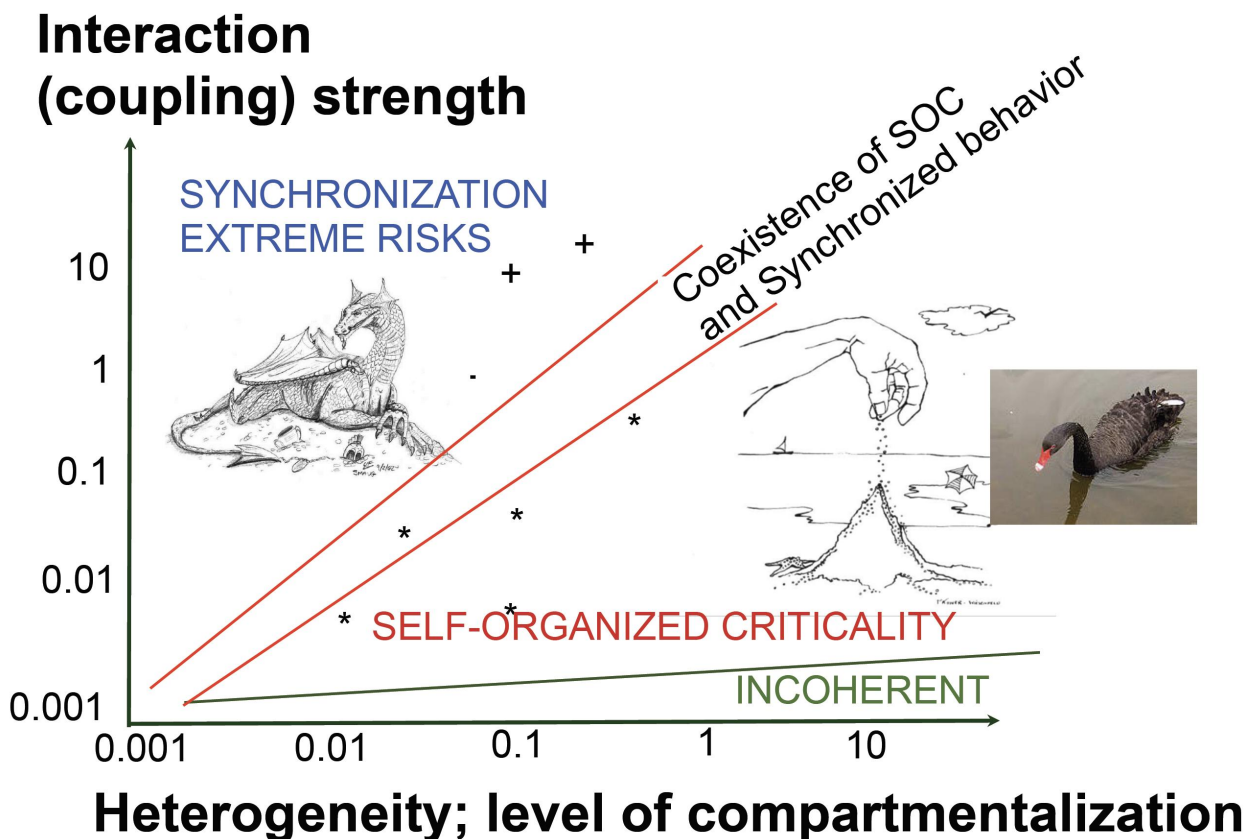
1.2 Potential mechanisms of dragon-king events

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- (1) Exogenous shocks
外源性冲击
- (2) Partial global synchronization
部分全局同步

coupling strength C
heterogeneity magnitude H



Courtesy of Prof. Sornette

1 Dragon-king events and dragon-king theory

1.2 Potential mechanisms of dragon-king events

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- (1) Exogenous shocks
外源性冲击
- (2) Partial global synchronization
部分全局同步
- (3) Clustering behavior and percolation phase transitions
聚簇行为与逾渗相变

Bose-Einstein condensate droplets

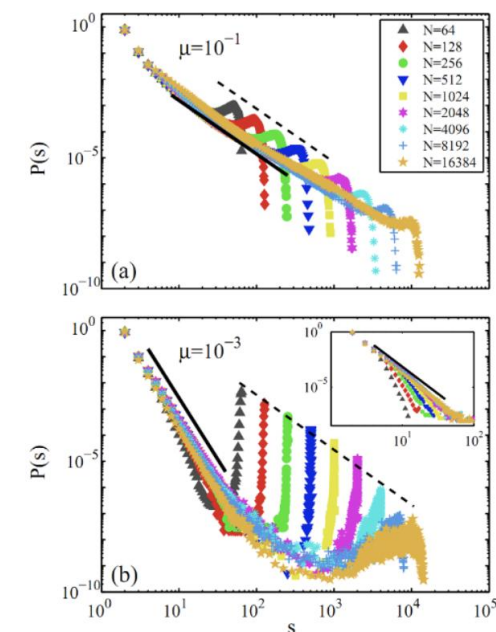
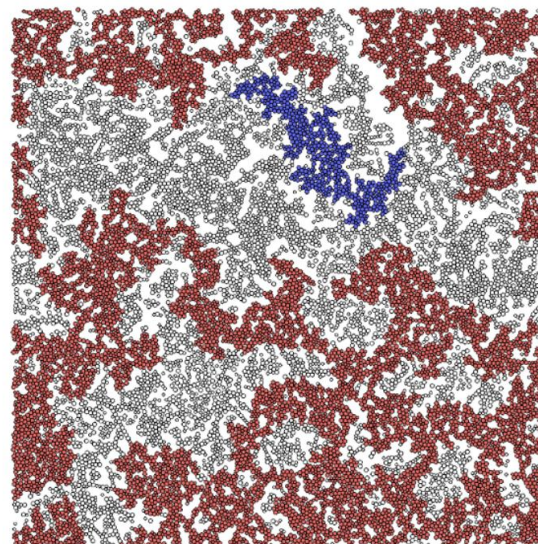
玻色-爱因斯坦凝聚滴模型

The attractiveness of city sizes

城市规模对人口吸引力

Grand canonical minority games

巨正则少数派博弈



Wang et al., 2020

1 Dragon-king events and dragon-king theory

1.2 Potential mechanisms of dragon-king events

Dragon-king events arise from unique amplification mechanisms that do not affect (or affect to lesser degree) other events in the same system, thus endowing dragon-king events with their inherent specific properties.

“龙王”事件的发生基于其独有的放大机制，该机制无法（或较少）作用于同一系统其他事件，因此赋予了“龙王”事件本质上的特殊属性。

- (1) Exogenous shocks

外源性冲击

- (2) Partial global synchronization

部分全局同步

- (3) Clustering behavior and percolation phase transitions

聚簇行为与逾渗相变

- (4) Positive feedback mechanism

正反馈机制

the Arctic ice melts

the value of a social media platform

financial markets change

running buffaloes

.....

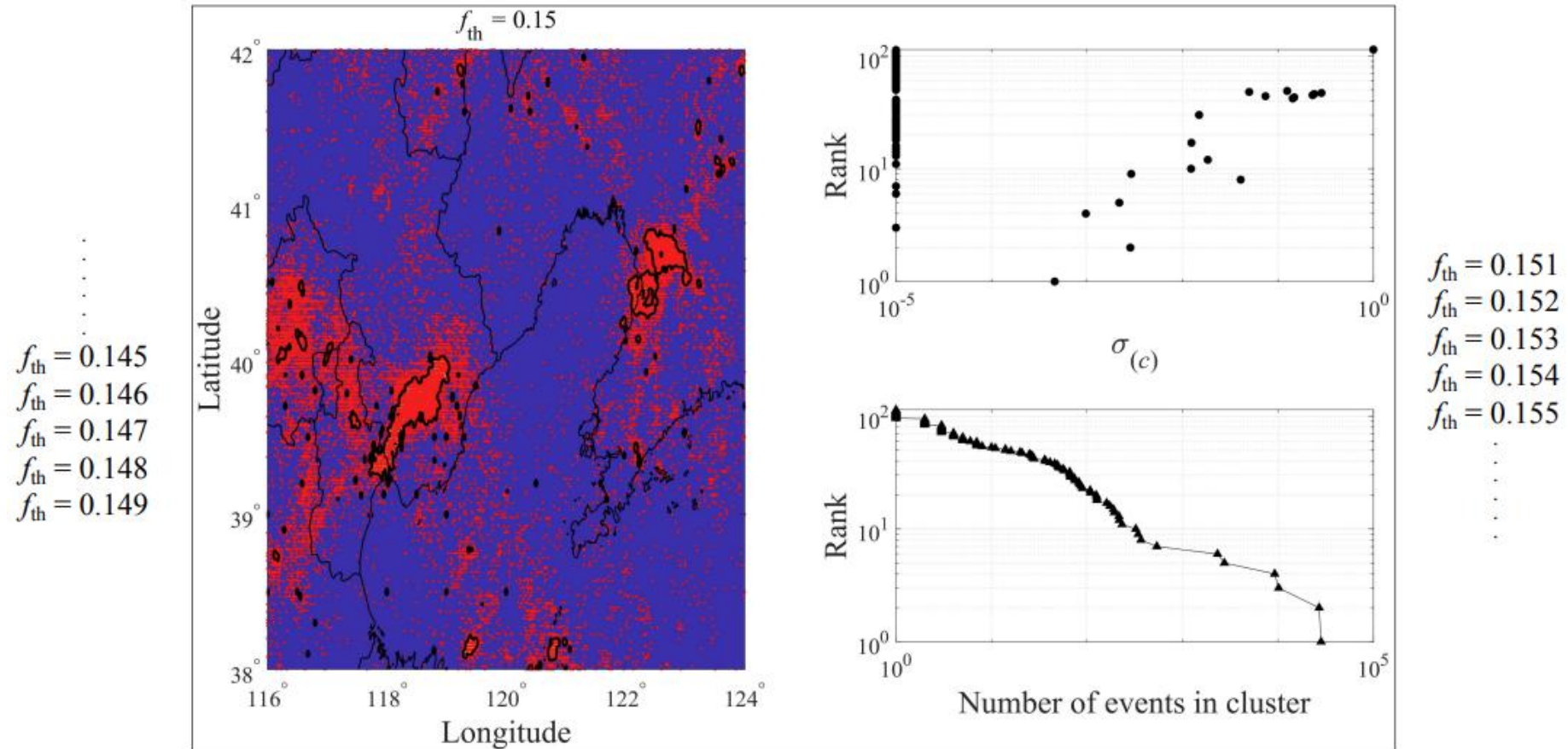
Positive feedback refers to a process where an initial change is amplified by the subsequent reactions it provokes, leading to a further increase in the effect (Sornette, 1999; Sammis and Sornette, 2002). This can create a self-reinforcing loop where the initial stimulus is continuously intensified. In other words, an initial deviation from some starting point is followed by increasing deviations.

正反馈指的是一种初始变化通过其引发的后续反应被放大，从而导致效果进一步增强的过程（Sornette, 1999; Sammis and Sornette, 2002）。这可能会产生一个初始刺激不断被加强的自我强化循环。换句话说，初始偏离某个起点后，随之而来的是不断增加的偏离。

2 Dragon-king v.s. Gutenberg-Richter

2.1 Defining the study region

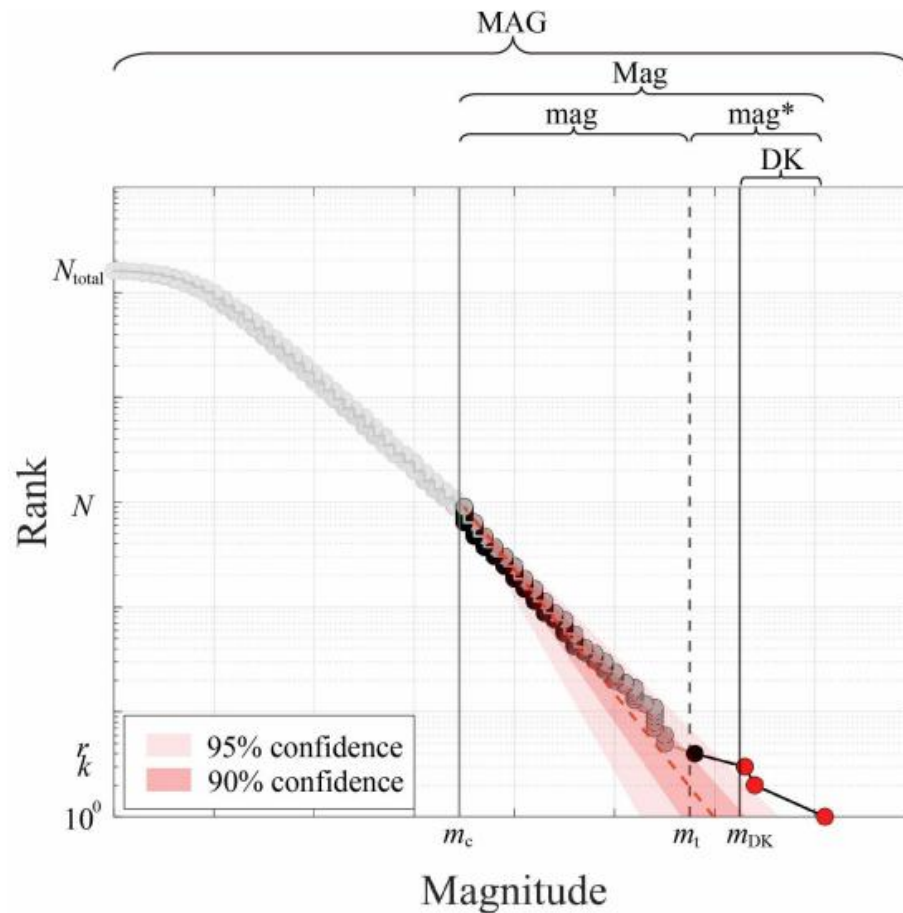
$$f_{[T_1, T_2]}(\text{Lon}, \text{Lat}) = \frac{1}{S_{\text{norm}}} \sum_{\substack{i=1 \\ t_i \in [T_1, T_2]}}^{N_{\text{raw}}} \frac{1}{2\pi h_i^2} \exp \left[\alpha m_i - \frac{(\text{Lon} - \text{Lon}_i^{\text{eq}})^2 + (\text{Lat} - \text{Lat}_i^{\text{eq}})^2}{2h_i^2} \right]$$



2 Dragon-king v.s. Gutenberg-Richter

2.2 Identifying dragon-king candidates

$$D^2(m_t; \text{Mag}) = -N - 2 \sum_{\substack{i=1 \\ m_{(i)} \leq m_t}}^N \left\{ \frac{2i}{2N+1} \log_{10} \left[\frac{\text{rank}(m_{(i)})}{N} \right] + \left(1 - \frac{2i}{2N+1} \right) \log_{10} \left[\frac{1 - \text{rank}(m_{(i)})}{N} \right] \right\}$$



2.3 Rigorous statistical testing

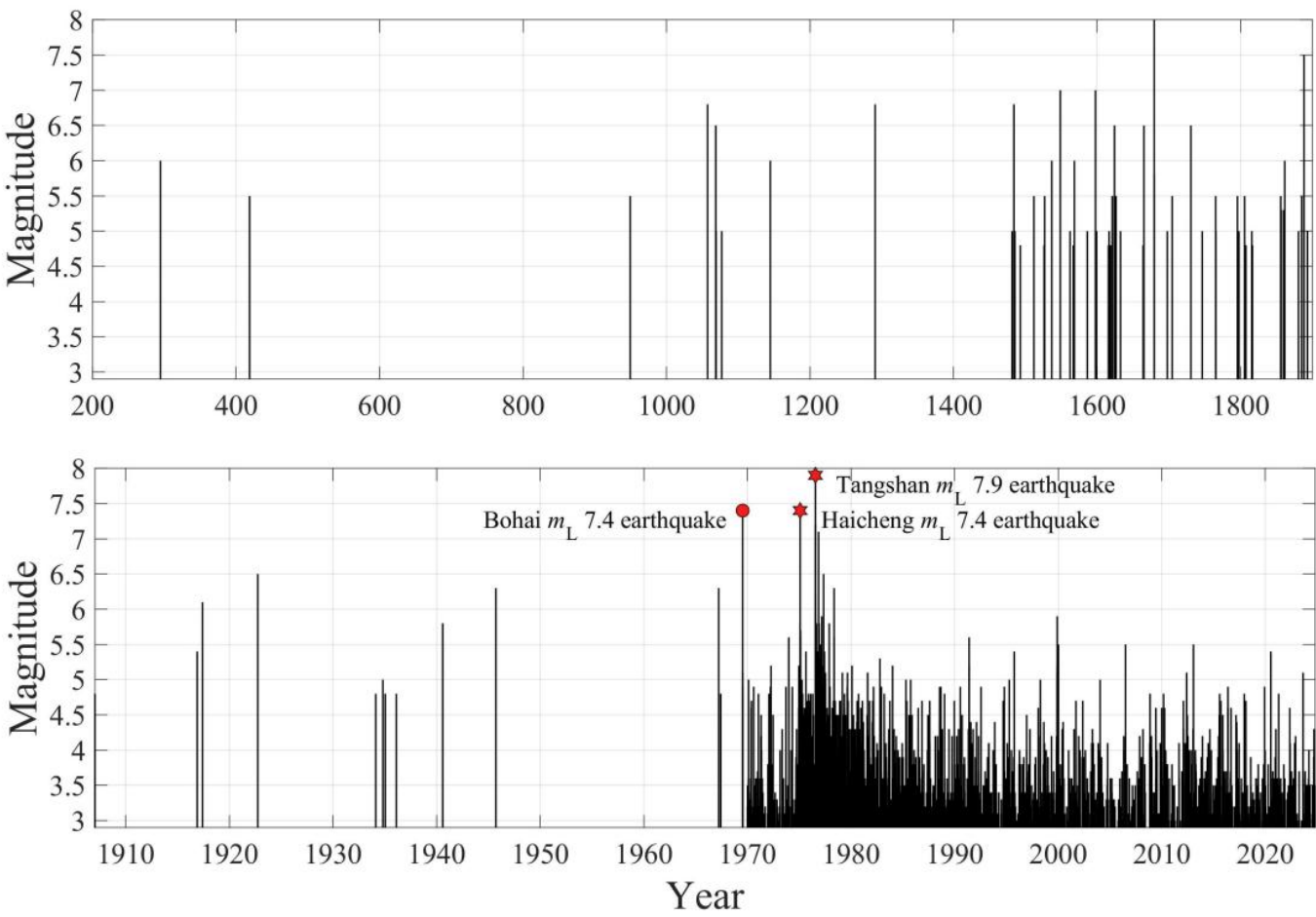
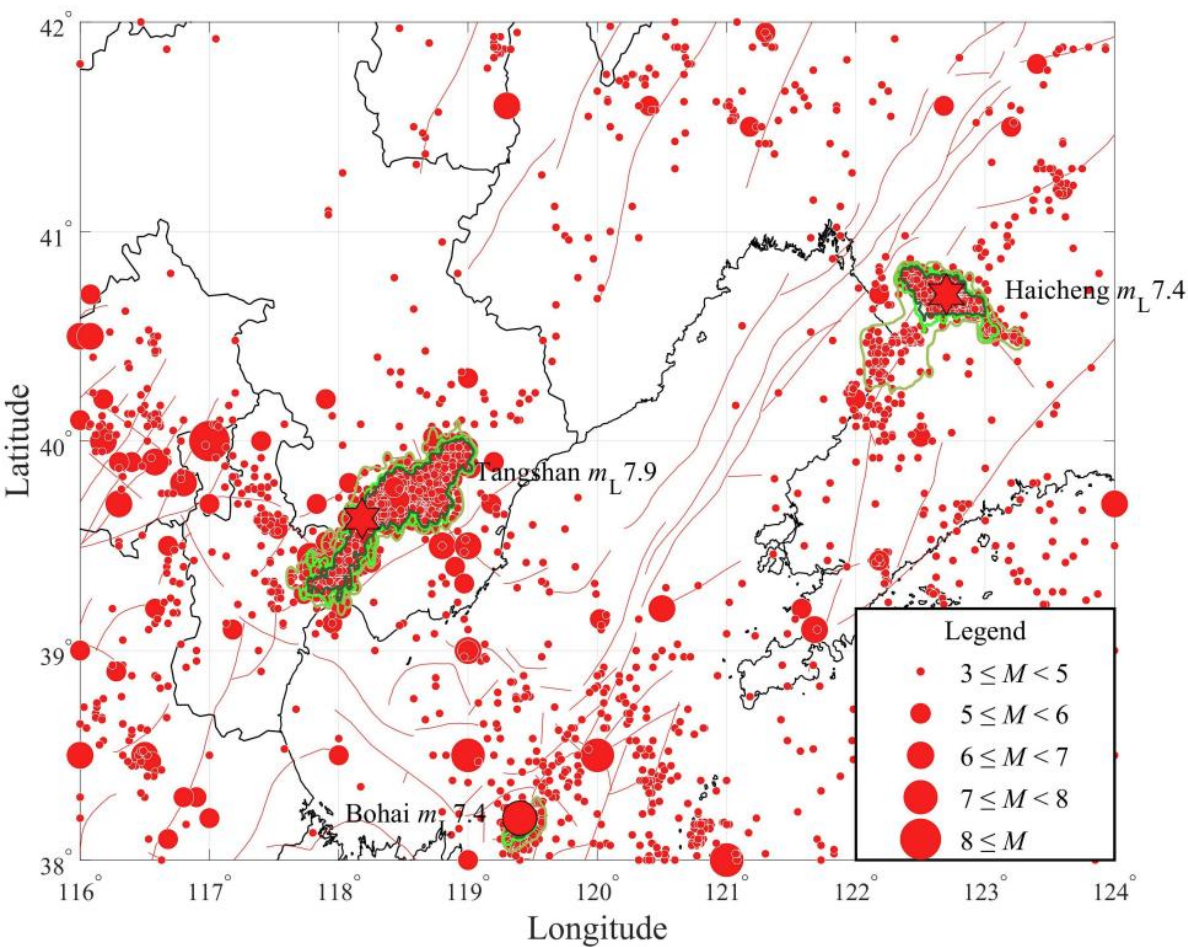
$$T_{j, N-v}^{\text{MRS}} = \frac{m_{(j)}}{\sum_{i=1}^{N-v} m_{(i)}}$$

2.4 Validating dragon-king earthquakes

$$T_{k, N-v}^{\text{SRS}} = \frac{\sum_{i=N-k+1}^N m_{(i)}}{\sum_{i=1}^{N-v} m_{(i)}}$$

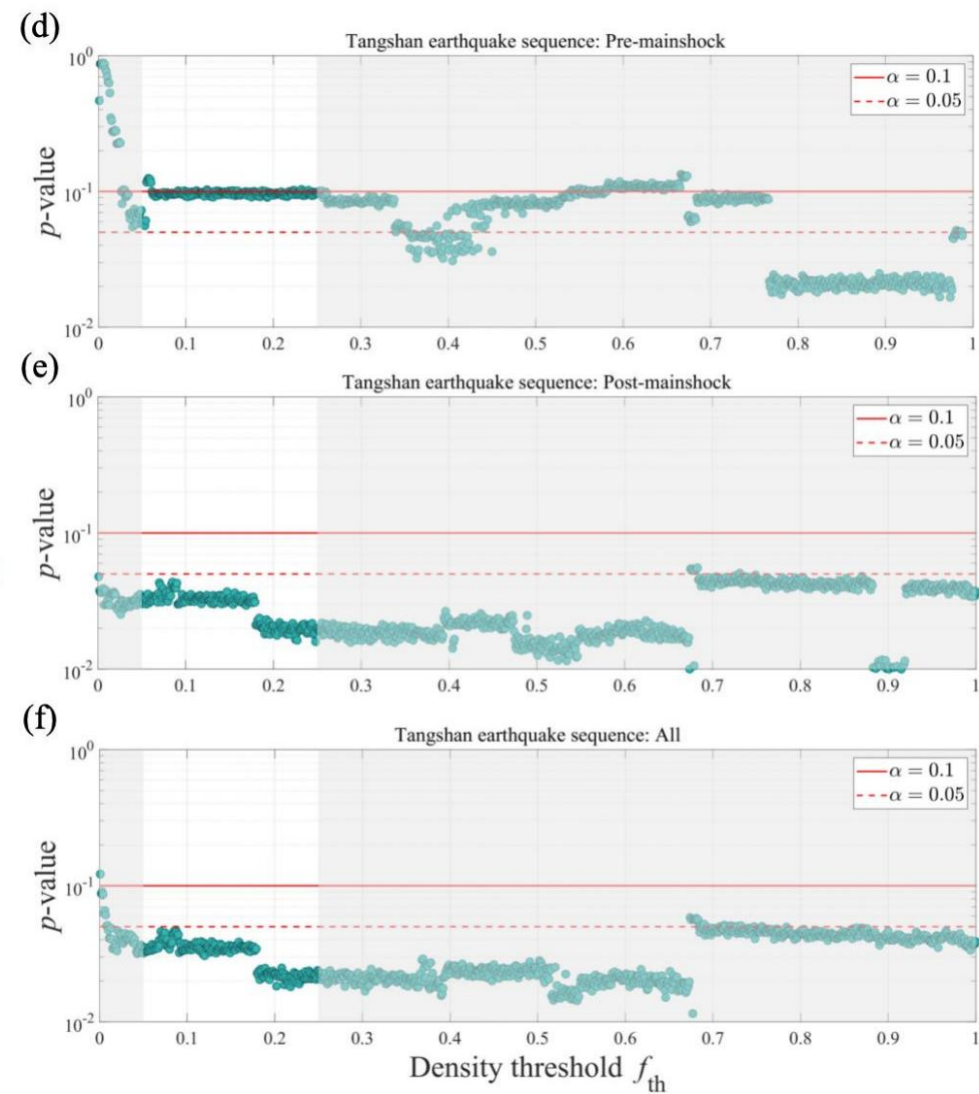
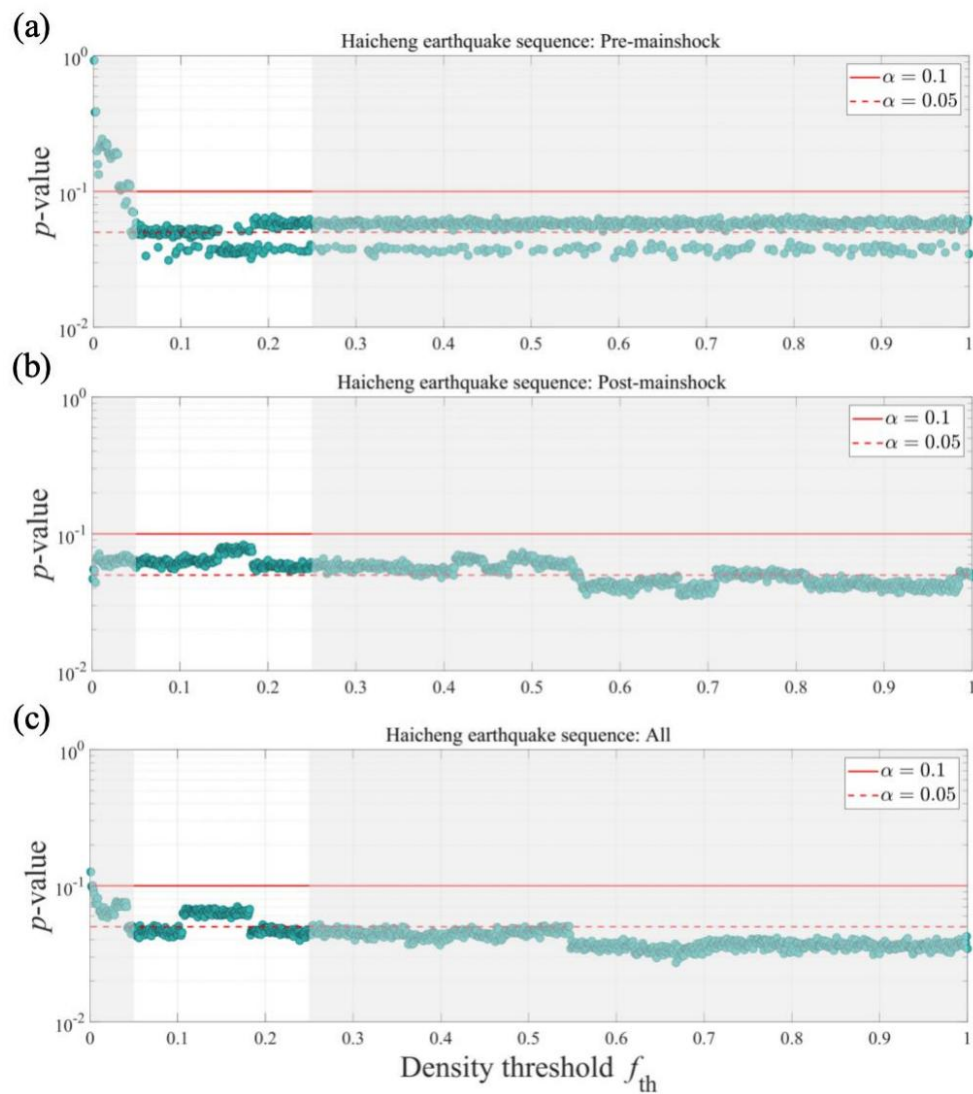
3 Dragon-king earthquakes?

3.1 Haicheng and Tangshan earthquakes



3 Dragon-king earthquakes?

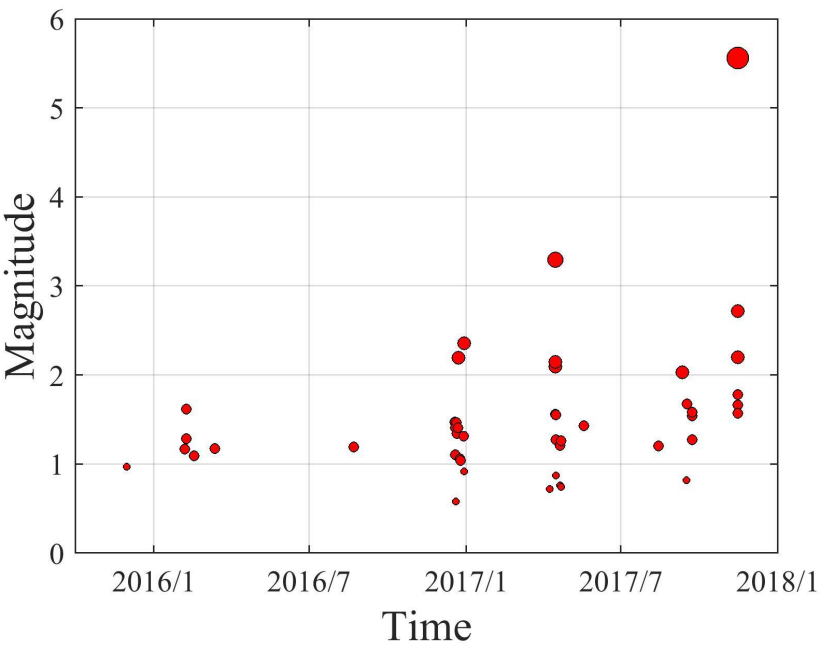
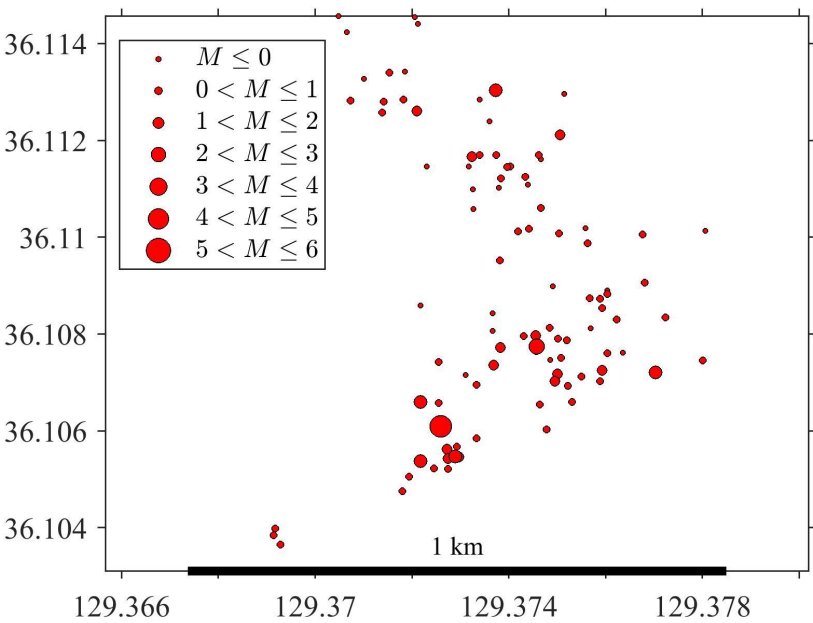
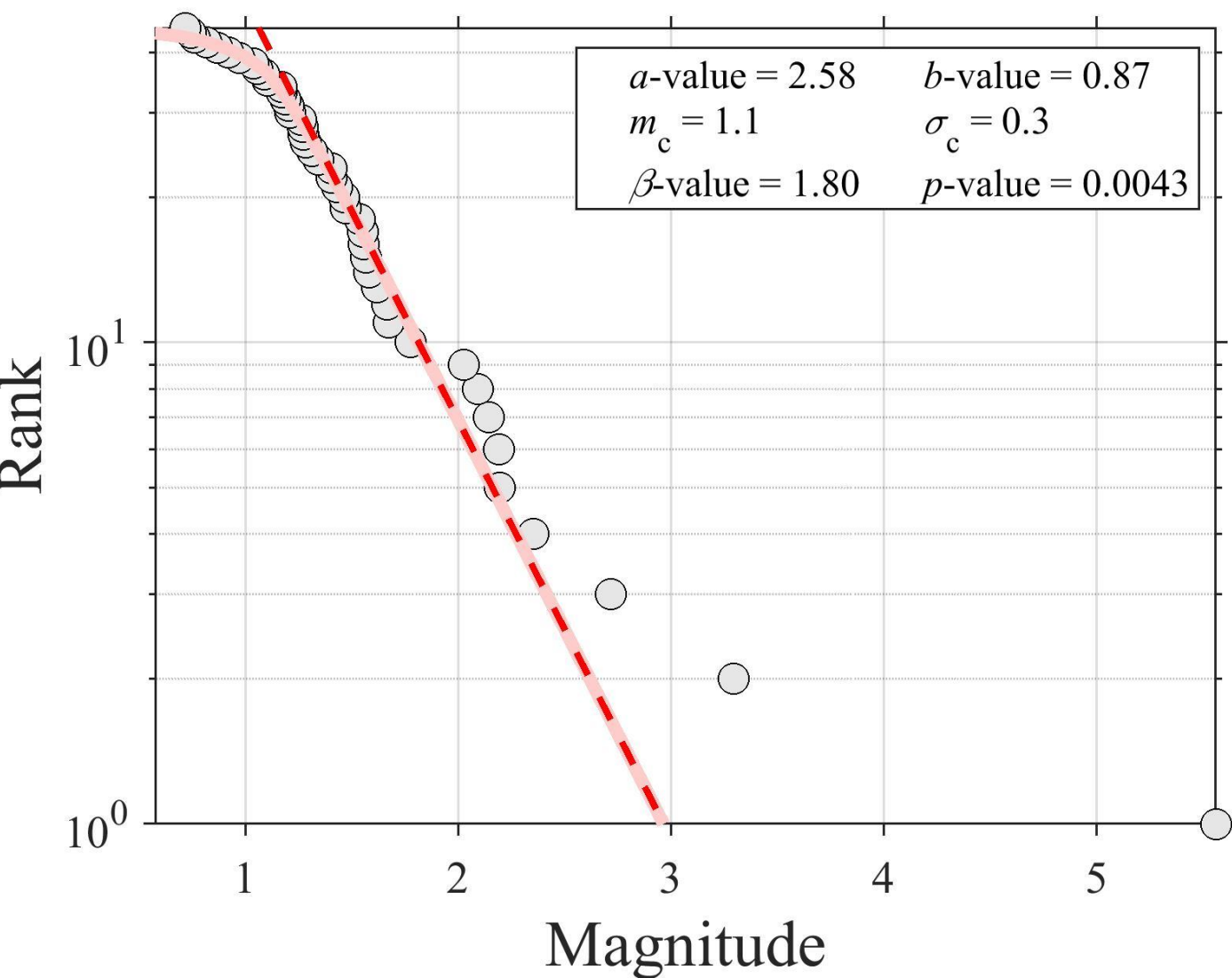
3.1 Haicheng and Tangshan earthquakes



3 Dragon-king earthquakes?

3.2 Injection-induced seismicity

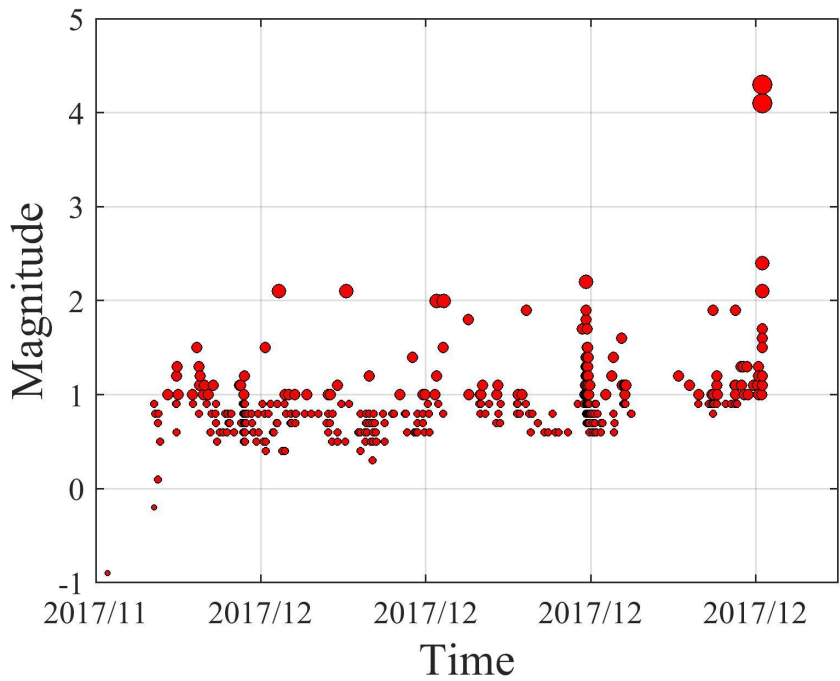
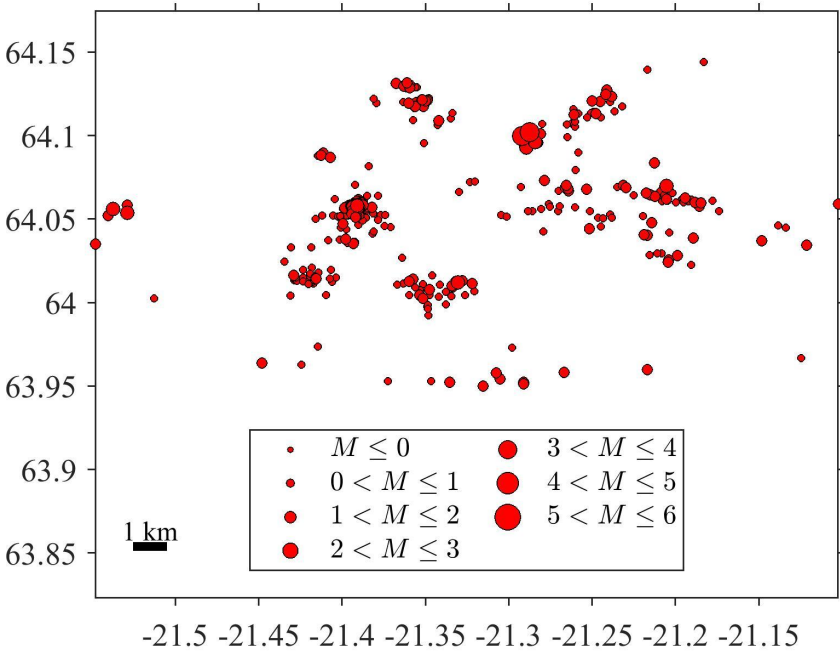
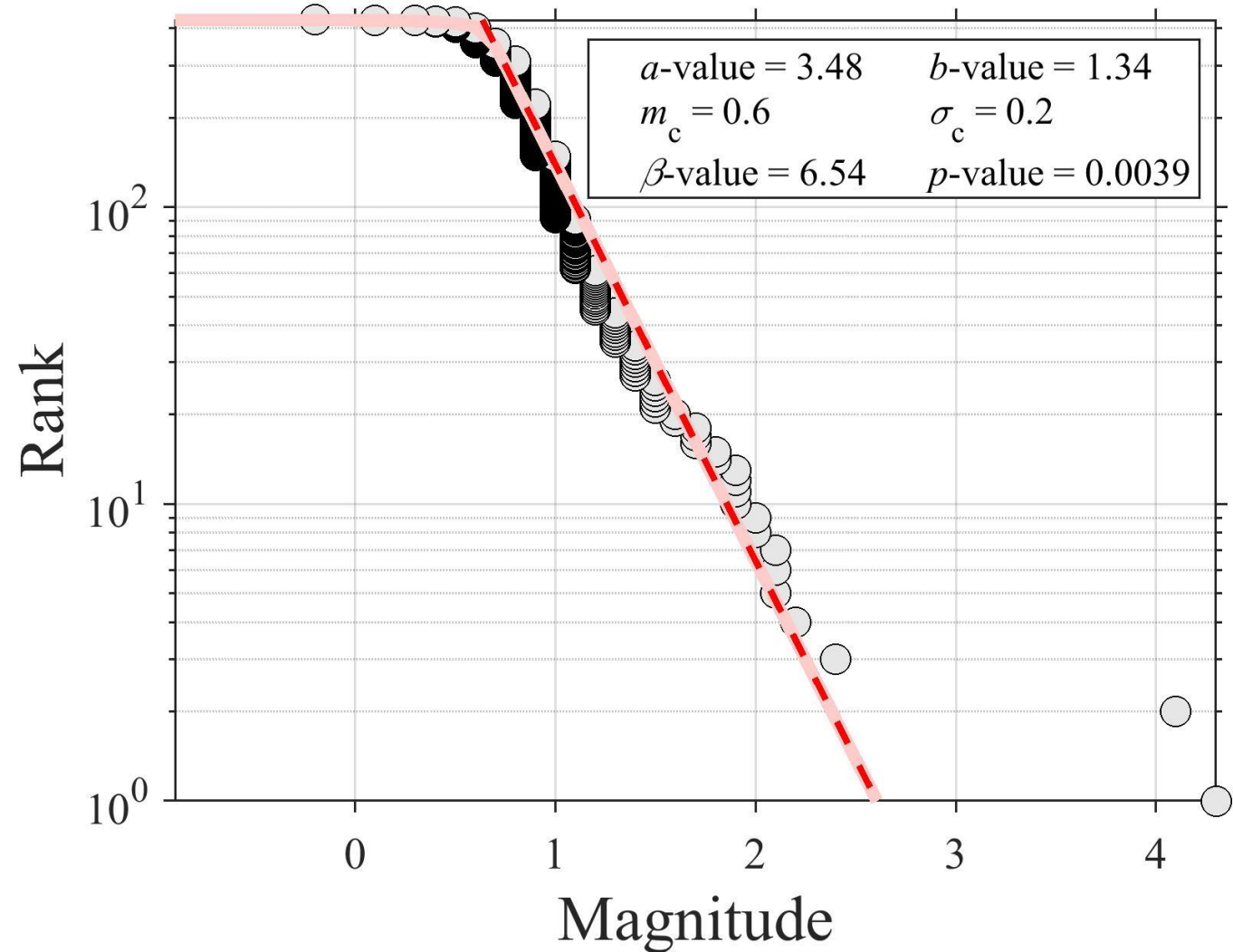
2017-11-15 05:29:31 M_w 5.5;
Pohang enhanced geothermal systems in Korea



3 Dragon-king earthquakes?

3.2 Injection-induced seismicity

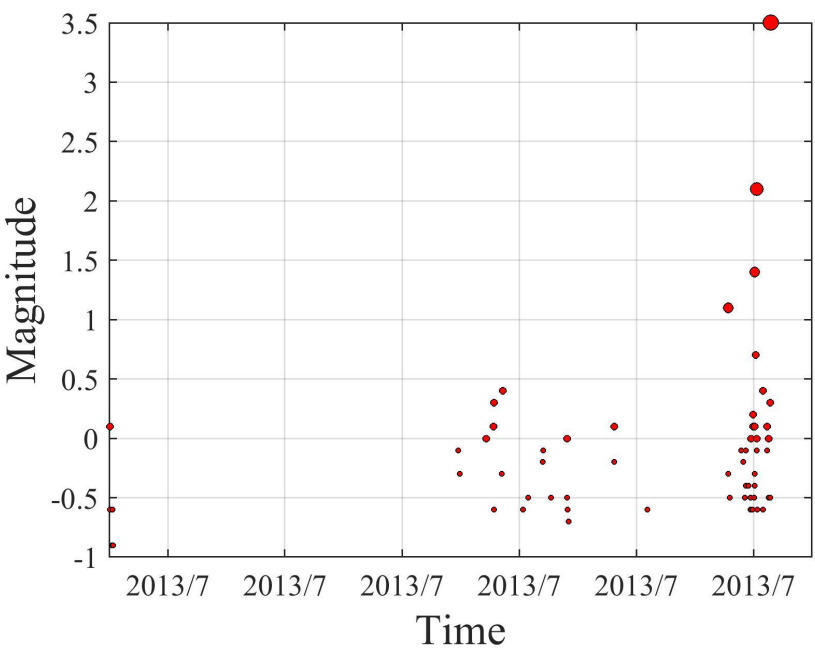
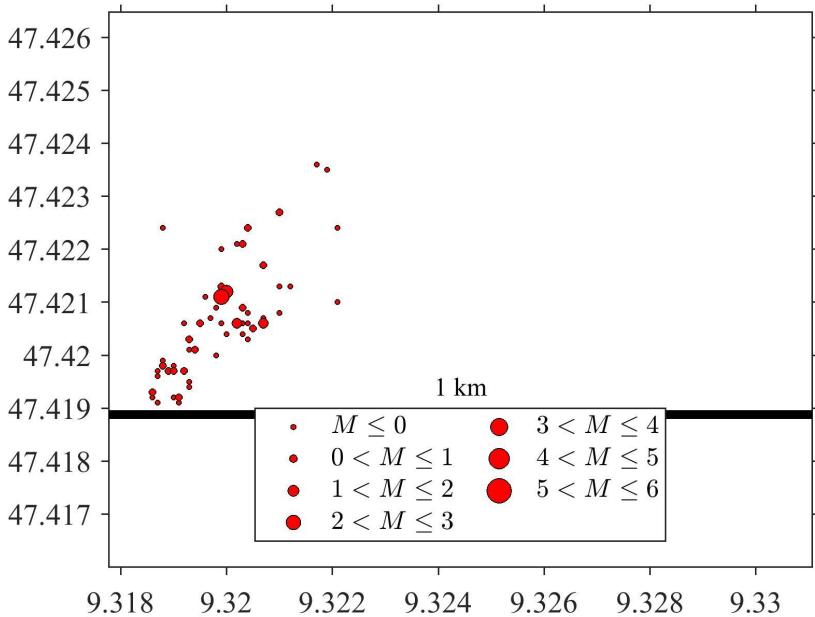
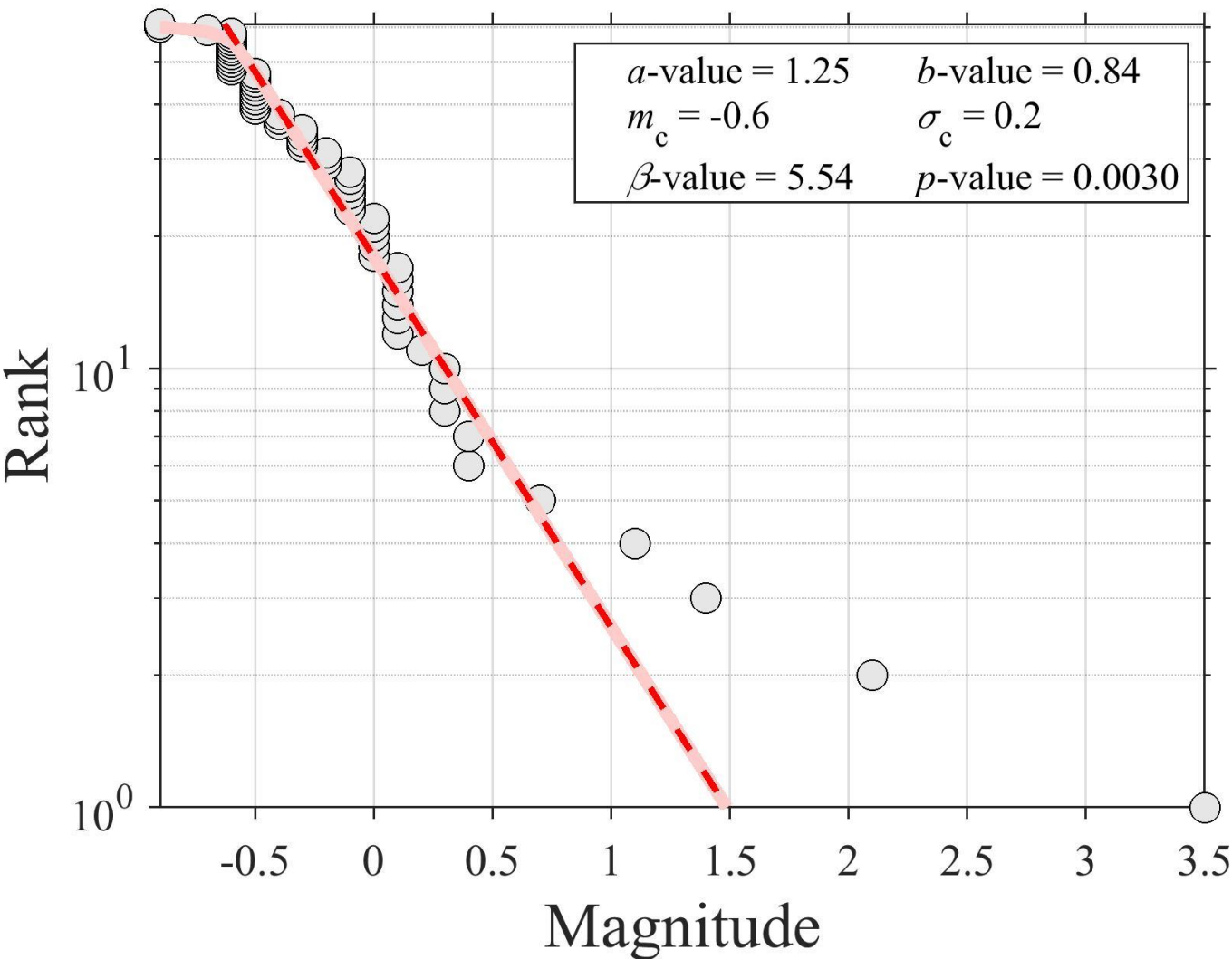
2017-12 M_w 4.2;
Geothermal energy production & Co2 sequestration; Carbfix, Iceland



3 Dragon-king earthquakes?

3.2 Injection-induced seismicity

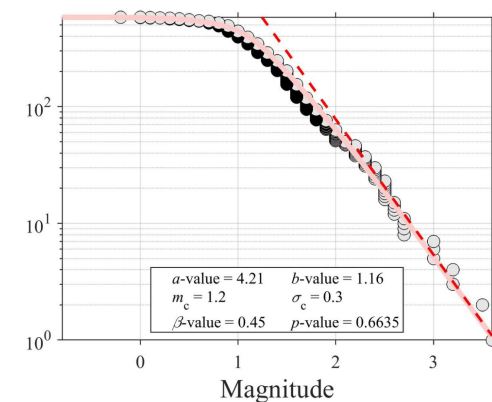
2013-7 M_w 3.5;
Enhanced geothermal systems in St. Gallen, Switzerland



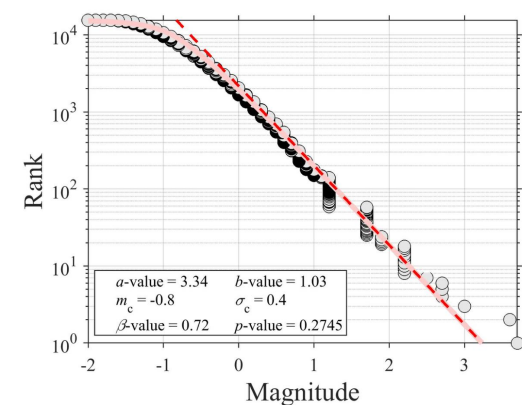
3 Dragon-king earthquakes?

3.2 Injection-induced seismicity

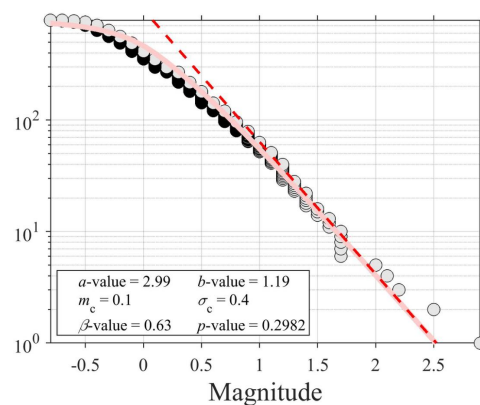
2013, NGF_NL_Groningen



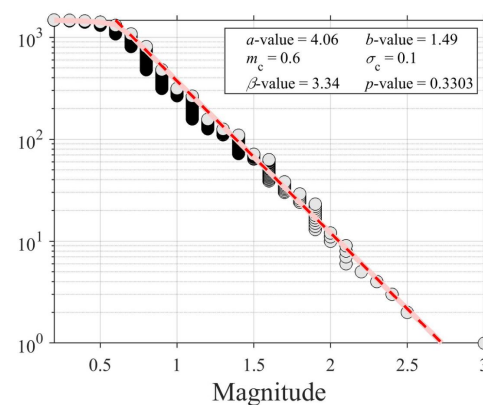
2003/12, EGS_AU_Cooper_Basin



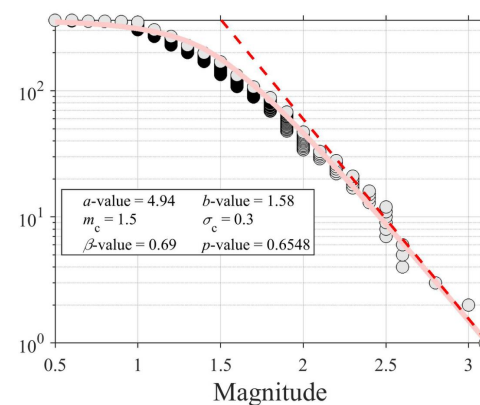
2005/9, EGS_AU_Cooper_Basin



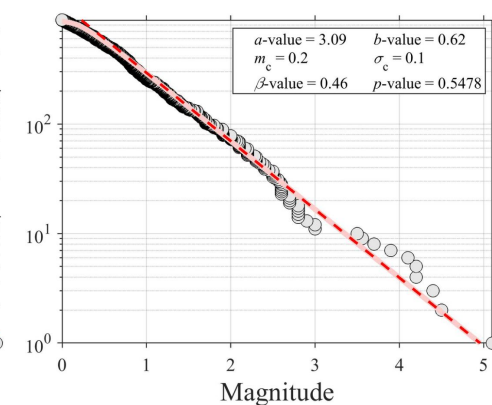
2006/12, EGS_CH_Basel



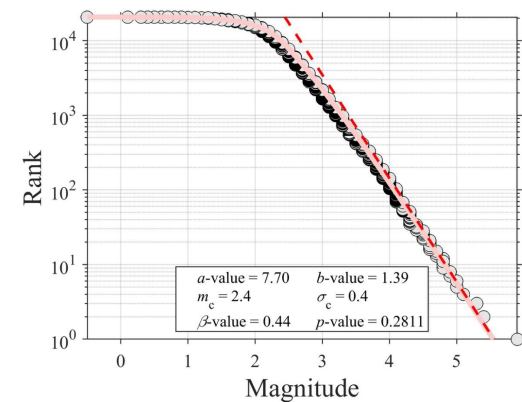
2007/9, EGS_US_Geysers



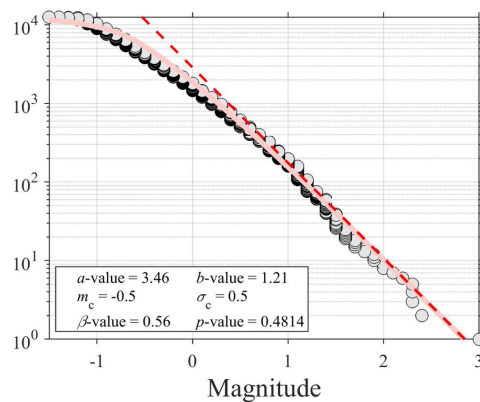
2010/7, NGF_CHN_Rongchang



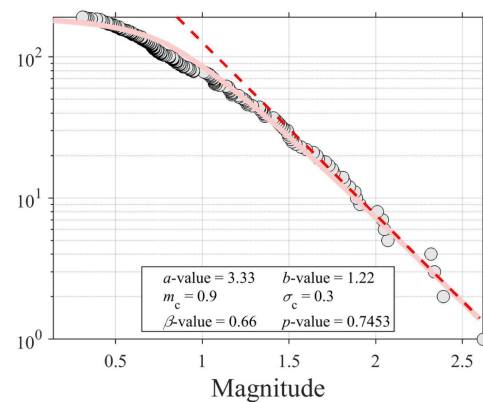
2016/6, Wastewater_US_Oklahoma



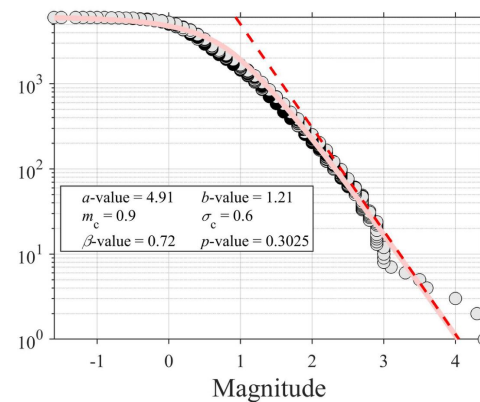
2012/11, EGS_AU_Cooper_Basin



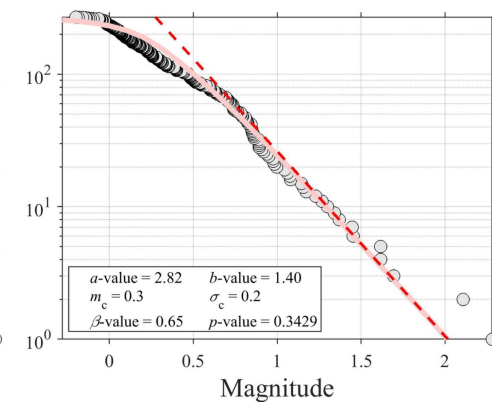
2012/12, EGS_US_Newberry



2013, Wastewater_US_ParadoxValley



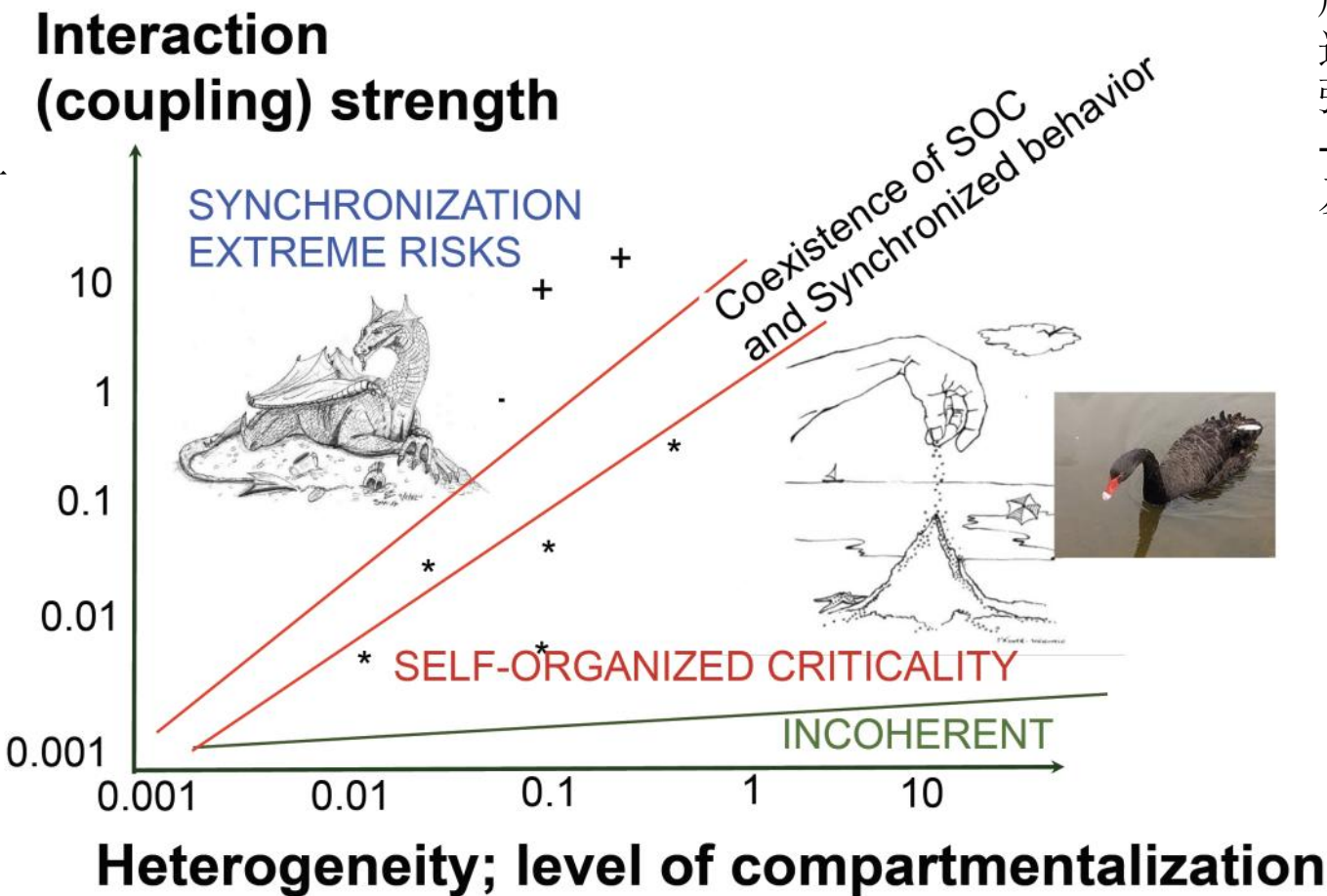
2011/11, EGS_US_Newberry



3 Dragon-king earthquakes?

3.2 Injection-induced seismicity

压力消散 → 有效正应力回升
矿物沉淀与裂缝愈合
颗粒致密化与应变硬化
摩擦记忆与速率-状态演化
多过程综合结果



当流体扩散-应力均化与化学-力学致密/愈合共同作用时，断层先经历“均匀化-弱化”再进入“再耦合-加固”阶段；强度增幅最终取决于注入体积-孔隙体积比、温度-化学体系及停注时机。

孔隙压扩散平滑有效法向应力
水力破裂连接裂缝网络
孔隙弹性-应力再分配

Take-home message

- ✓ **Using adaptive kernel density estimation, we propose an objective clustering method that partitions the seismic field into distinct regions.**
- ✓ **We develop a framework to test the dragon-king earthquake hypothesis, using spatial clustering and a high-power outlier detection technique.**
- ✓ **Dragon-king signatures in the Haicheng and Tangshan main shocks suggest some large and great quakes mature in ways that boost predictability.**

Li, J., & Sornette, D. (2025). Are Haicheng and Tangshan Earthquakes Dragon-Kings?. (under review) or arXiv preprint arXiv:2504.21310.

Li, J., Sornette, D., Wu, Z., Li, H. (2025). New horizon in the statistical physics of earthquakes: Dragon-king theory and dragon-king earthquakes. Reviews of Geophysics and Planetary Physics, 56(3): 225-242 (in Chinese). doi:10.19975/j.dqyxx.2024-033.

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Thank you!